

Exploratory Data Analysis (EDA)

Task 2 - EDA Report

DecodeLabs Internship- Data Analytics

Batch MAY 5 - JUNE 5 2026

Analyst	Dataset	Date	Tool
Sahaji Perera	Dataset_for_Data_Analytics.xlsx (Cleaned)	2026-05-17	Microsoft Excel
Input Source	Total Records	Date Range	Status
Week 1 Cleaned Output	1,200 rows 14 columns	2023-01-01 to 2025-06-30	All Phases Complete

Phase 1 -Dataset Overview

1.1 Column Inventory

The cleaned dataset contains 14 columns classified into three analytical categories:

Column	Data Type	Category	Description
OrderID	Text	Identifier	Unique order reference
Date	Date	Identifier	Order date (2023-01 to 2025-06)
CustomerID	Text	Identifier	Unique customer reference
Product	Text	Categorical	Product purchased (7 types)
Quantity	Integer	Numerical	Units ordered (1–5)
UnitPrice	Decimal	Numerical	Price per unit (\$11.39–\$699.93)
ShippingAddress	Text	Identifier	Delivery address
PaymentMethod	Text	Categorical	Payment type (5 types)
OrderStatus	Text	Categorical	Order state (5 types)
TrackingNumber	Text	Identifier	Shipment tracking code
ItemsInCart	Integer	Numerical	Items browsed before ordering (1–10)
CouponCode	Text	Categorical	Discount code used (4 types)
ReferralSource	Text	Categorical	Acquisition channel (5 types)
TotalPrice	Decimal	Numerical	Final order value (\$11.39–\$3,456.40)

Analytical scope: 4 Numerical columns | 5 Categorical columns | 5 Identifier columns (excluded from statistics)

Phase 2 - Descriptive Statistics

2.1 Summary Statistics -All Numerical Columns

Statistic	Quantity	UnitPrice	ItemsInCart	TotalPrice
COUNT	1,200	1,200	1,200	1,200
MEAN	2.95	\$356.41	5.49	\$1,053.97
MEDIAN	3.00	\$364.21	5.00	\$823.62
MIN	1.00	\$11.39	1.00	\$11.39
MAX	5.00	\$699.93	10.00	\$3,456.40
STD DEV	1.41	\$197.18	2.28	\$819.86
Q1 (25%)	2.00	\$186.06	4.00	\$410.52
Q3 (75%)	4.00	\$521.57	7.00	\$1,578.47
SKEWNESS	0.028	-0.027	0.001	0.891 Δ

2.2 Skewness Interpretation

Column	Skew	Shape	Business Meaning
Quantity	0.028	Symmetric	Evenly distributed 1–5; no unusual ordering patterns
UnitPrice	-0.027	Symmetric	Prices balanced across full range; no price concentration
ItemsInCart	0.001	Symmetric	Balanced browsing behaviour before purchase
TotalPrice	0.891	Right-Skewed Δ	Mean (\$1,054) is \$230 above median (\$824); a small number of large bulk orders inflate the average — use median as the true typical order value

Phase 3 - Categorical Frequency Analysis

3.1 Product Distribution

Product	Order Count	% of Total
Printer	181	15.08%
Tablet	179	14.92%
Chair	178	14.83%
Laptop	173	14.42%
Desk	170	14.17%
Monitor	163	13.58%
Phone	156	13.00%
TOTAL	1,200	100.00%

Observation: Products are almost perfectly balanced (181 to 156 orders). No single product dominates, indicating healthy inventory spread. Difference between highest and lowest is only 25 orders.

3.2 Payment Method Distribution

Payment Method	Order Count	% of Total
Online	258	21.50%
Cash	246	20.50%
Credit Card	234	19.50%
Debit Card	232	19.33%
Gift Card	230	19.17%
TOTAL	1,200	100.00%

Observation: All 5 payment methods are used almost equally (~20% each). Online is slightly preferred. No single payment method dominates.

3.3 Order Status Distribution **⚠ Critical Finding**

Order Status	Order Count	% of Total
Cancelled	250	20.83%
Returned	247	20.58%
Pending	237	19.75%
Shipped	235	19.58%
Delivered	231	19.25%
TOTAL	1,200	100.00%

CRITICAL: $250 + 247 = 497$ orders (41.4%) are lost to cancellation or return. Only 231 orders (19.2%) are successfully delivered — less than 1 in 5. This is the single most urgent business problem identified in this dataset.

3.4 Coupon Code Distribution

Coupon Code	Order Count	% of Total
FREESHIP	313	26.08%
NO_COUPON	309	25.75%
WINTER15	292	24.33%
SAVE10	286	23.83%
TOTAL	1,200	100.00%

Observation: 891 of 1,200 customers (74.2%) used a coupon. FREESHIP is the most popular. All three coupon types are balanced in usage.

3.5 Referral Source Distribution

Referral Source	Order Count	% of Total
Instagram	259	21.58%
Email	250	20.83%
Google	241	20.08%

Facebook	228	19.00%
Referral	222	18.50%
TOTAL	1,200	100.00%

Observation: Instagram drives the most traffic (259 orders). Word-of-mouth Referral brings the fewest (222 orders).

Phase 4 -Outlier Detection (IQR Method)

4.1 IQR Calculation Summary

Calculation	Quantity	UnitPrice	ItemsInCart	TotalPrice
Q1	2.00	\$186.06	4.00	\$410.52
Q3	4.00	\$521.57	7.00	\$1,578.47
IQR	2.00	\$335.51	3.00	\$1,167.95
Lower Fence	-1.00	-\$317.20	-0.50	-\$1,341.41
Upper Fence	7.00	\$1,024.83	11.50	\$3,330.41
Outliers Found	0	0	0	8 

4.2 The 8 Flagged Outlier Orders

OrderID	Product	Qty	UnitPrice	TotalPrice	Status	Payment	Source
ORD200107	Printer	5	\$670.75	\$3,353.75	Shipped	Gift Card	Instagram
ORD200326	Laptop	5	\$670.48	\$3,352.40	Returned	Gift Card	Facebook
ORD200328	Tablet	5	\$674.04	\$3,370.20	Cancelled	Online	Google
ORD200469	Chair	5	\$676.98	\$3,384.90	Cancelled	Cash	Facebook
ORD200632	Laptop	5	\$678.16	\$3,390.80	Delivered	Gift Card	Facebook
ORD200789	Tablet	5	\$691.28	\$3,456.40	Delivered	Online	Email
ORD201065	Printer	5	\$666.80	\$3,334.00	Delivered	Debit Card	Referral
ORD201122	Monitor	5	\$678.19	\$3,390.95	Returned	Online	Facebook

Pattern: ALL 8 outliers share the same profile — Quantity = 5 (maximum) with UnitPrice above \$666. These are legitimate bulk purchases of premium-priced items, not data entry errors. Decision: Keep all 8 records. Flag for VIP customer investigation. Note: 3 of the 8 outlier orders were Cancelled or Returned.

Phase 5 -Group & Revenue Analysis

5.1 Revenue by Product

Product	Total Revenue	Avg Order Value	Orders	Bad Rate
Chair	\$195,620	\$1,098.99	178	41.0%
Printer	\$195,613	\$1,080.73	181	40.3%

Laptop	\$192,127	\$1,110.56	173	42.8%
Tablet	\$186,569	\$1,042.28	179	43.0%
Monitor	\$175,651	\$1,077.62	163	43.6% Δ
Desk	\$167,460	\$985.06	170	39.4%
Phone	\$151,722	\$972.58	156	39.7%
TOTAL	\$1,264,762	—	1,200	41.4%

5.2 Revenue by Referral Source

Source	Total Revenue	Avg Order Value	Orders	Rank
Instagram	\$275,285	\$1,062.88	259	#1 Volume
Email	\$261,809	\$1,047.23	250	#2
Google	\$250,441	\$1,039.18	241	#3
Facebook	\$250,411	\$1,098.29	228	#1 Avg Value
Referral	\$226,816	\$1,021.69	222	#5

5.3 Revenue by Payment Method

Payment Method	Orders	Total Revenue	Avg Order Value
Credit Card	234	\$263,848	\$1,127.55
Online	258	\$262,443	\$1,017.22
Cash	246	\$259,786	\$1,056.04
Gift Card	230	\$246,324	\$1,070.97
Debit Card	232	\$232,361	\$1,001.56

5.4 Coupon Code Revenue Analysis

Coupon	Orders	Total Revenue	Avg Order	Median Order
FREESHIP	313	\$335,037	\$1,070.41	\$793.40
NO_COUPON	309	\$322,401	\$1,043.37	\$834.12
SAVE10	286	\$304,840	\$1,065.87	\$823.06
WINTER15	292	\$302,484	\$1,035.90	\$838.71

Phase 6 -Trend Analysis

6.1 Monthly Revenue Highlights (Jan 2023 – Jun 2025)

Metric	Month	Value
Peak Revenue Month	June 2024	\$68,068.54 (53 orders)
2nd Highest Revenue	May 2023	\$63,836.84 (49 orders)
Lowest Revenue Month	April 2023	\$27,751.71 (31 orders)
Lowest Order Volume	January 2025	27 orders (\$29,099)
Dataset Span	30 months	Jan 2023 – Jun 2025

Pattern: Revenue fluctuates significantly month to month with no consistent upward trend. Spikes in May 2023 and June 2024 suggest seasonal or promotional events. The business has not shown organic growth over the 30-month period.

Phase 7- Correlation Analysis

7.1 Correlation Matrix

	Quantity	UnitPrice	ItemsInCart	TotalPrice
Quantity	1.000	0.015	0.650	0.615
UnitPrice	0.015	1.000	0.001	0.717
ItemsInCart	0.650	0.001	1.000	0.393
TotalPrice	0.615	0.717	0.393	1.000

7.2 Correlation Interpretation

Variable Pair	r Value	Strength	Business Meaning
UnitPrice ↔ TotalPrice	0.717	Strong	Higher-priced products generate significantly more revenue per order
Quantity ↔ TotalPrice	0.615	Moderate	Ordering more units increases order value but less powerfully than price
Quantity ↔ ItemsInCart	0.650	Moderate	Customers who browse more items tend to order more units
ItemsInCart ↔ TotalPrice	0.393	Weak-Moderate	Browsing more items does not strongly predict higher spending
UnitPrice ↔ Quantity	0.015	Negligible	Expensive products are not bought in lower quantities than cheap ones
UnitPrice ↔ ItemsInCart	0.001	Negligible	Product price has no relationship to browsing behaviour

Important: Correlation does not equal causation. These relationships are clues to investigate, not proven cause-and-effect conclusions.

Phase 8 -Key Observations & Business Insights

ID	Area	What the Data Says	Business Diagnosis
OBS-01	Order Status	497 orders (41.4%) lost to Cancellation or Return. Only 231 (19.2%) Delivered.	Less than 1 in 5 orders reaches the customer. This is the single most urgent issue — requires immediate investigation by product, channel, and time period.
OBS-02	TotalPrice Skew	Mean \$1,054 vs Median \$824 — gap of \$230. Skewness = 0.891.	8 bulk outlier orders inflate the mean. Use median (\$823.62) as the true typical order value for pricing and benchmarking decisions.

OBS-03	Product Revenue	Laptop avg order = \$1,110.56 (highest). Phone avg = \$972.58 (lowest).	Laptops generate the most value per transaction. Promoting premium laptops over high-volume phone sales yields better revenue per order.
OBS-04	Monitor Failure Rate	71 of 163 Monitor orders fail (43.6% bad rate — worst of all products).	Nearly every other Monitor order fails. Likely product quality, listing accuracy, or pricing issue specific to monitors. Requires urgent product investigation.
OBS-05	Coupon Impact	NO_COUPON median = \$834.12 (highest). FREESHIP median = \$793.40 (lowest).	74.2% of customers use coupons but spend less than non-coupon users. The business may be eroding margin unnecessarily on customers who would buy anyway.
OBS-06	Referral Channels	Instagram: 259 orders, \$275,285 revenue. Facebook: \$1,098.29 avg (highest).	Instagram best for volume; Facebook attracts higher-value customers. Recommend using Instagram for reach and Facebook for premium product promotion.
OBS-07	Revenue Driver	UnitPrice vs TotalPrice correlation = 0.717 (strongest in dataset).	Product price matters more than quantity sold. Stocking and promoting higher-priced items has greater revenue impact than running bulk-buy promotions.

Verification Gate - Pass Criteria

Phase	Requirement	Result	Pass/Fail
Phase 1-Overview	14 columns documented	14 columns inventoried	PASS
Phase 2-Descriptive Stats	9 statistics × 4 columns	All calculated + skew interpreted	PASS
Phase 3- Frequency	5 categorical columns analysed	All 5 pivot tables complete	PASS
Phase 4 -Outliers	IQR applied to all numericals	8 outliers found in TotalPrice	PASS
Phase 5 -Revenue Analysis	Multi-dimensional group analysis	Product, Referral, Payment, Coupon	PASS
Phase 6 -Trends	Monthly revenue trend	30-month trend identified	PASS
Phase 7-Correlation	Correlation matrix completed	All 6 pairs interpreted	PASS
Phase 8 - Insights	7 business observations	All 7 with So What? diagnosis	PASS